

Data Quality over Scale: A 30M-Parameter Encoder for Arabic Diacritization

Interscript ML Team

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Abstract

Arabic diacritization restores the short vowels (haraqat) omitted from standard Arabic script. The strongest published system, Sadeed, reaches 1.2% diacritization error rate (DER) using a 1.5B-parameter decoder-only language model. We show that a character-level Transformer encoder of roughly 30M parameters—two orders of magnitude smaller—reaches 0.99% DER when trained on a large, aggressively cleaned corpus. Our combined corpus merges the full 75M-word Tashkeela dump (cleaned with a Sadeed-style pipeline), Arabic Wikipedia, and QCRI EMNLP-2025 data, totalling 2.1M sentences. Scaling training data $28\times$ ($75\text{K} \rightarrow 2.1\text{M}$) reduced DER from 2.42% to 0.99%. We argue that for well-resourced morphological restoration tasks, data curation dominates model capacity, and we release all training and evaluation code.

1 Introduction

Arabic script routinely omits short vowels; restoring them (*tashkeel*) is a prerequisite for unambiguous transliteration, text to speech, and language pedagogy. Prior work has escalated model size: Sadeed [1] fine-tunes a 1.5B-parameter Arabic language model and reports 1.2% DER on SadeedDiac-25.

We take the opposite direction. Our contributions:

1. A compact char-level encoder (6 layers, $d = 384$, 6 heads, $\approx 30\text{M}$ parameters) that reaches **0.99% DER** on a held-out split of a 2.1M-sentence corpus.
2. A data ablation showing DER falls $2.42\% \rightarrow 0.99\%$ as the corpus grows $75\text{K} \rightarrow 2.1\text{M}$ ($28\times$).
3. An explicit evaluation-protocol analysis: our test split differs from SadeedDiac-25, and we state the comparison caveats.

2 Related Work

Sadeed [1] is the strongest published Arabic diacritizer (1.2% DER, 1.5B params). Earlier systems include Shakkelha, Tafran, and CTC-based encoders; all are dominated by both Sadeed and this work.

3 Data

3.1 Sources

- **Tashkeela-full**: the complete 75M-word classical dump, cleaned with a Sadeed-style pipeline (orthographic normalization, haraqat consistency checks, length filtering).
- **Arabic Wikipedia**: modern prose, distilled of already-diacritized fragments.
- **QCRI EMNLP-2025**: news-domain diacritized data.

Combined: 2.1M sentences (train), 240K (val), 52,224 held-out test.

3.2 Cleaning

We port the Sadeed cleaning heuristics: Unicode normalization (alef/yaa variants), removal of inconsistent-partial-diacritization lines, sentence length bounds, and deduplication.

4 Model

Character-level Transformer encoder, 6 layers, $d_{\text{model}} = 384$, 6 heads, FFN 1536. Input is the undiacritized character sequence; each consonant position emits a multi-class head over haraqat combinations. Training: cross-entropy with label smoothing 0.1, cosine schedule, 15 epochs, batch 64.

5 Experiments

5.1 Data scaling

Corpus	Sentences	DER
Initial (Tashkeela sample)	75K	2.42%
Combined v2 (full)	2.1M	0.99%

Table 1: $28\times$ data scaling halves the error rate.

5.2 Comparison with published SOTA

System	Params	DER	Test set
Sadeed (published)	1.5B	1.2%	SadeedDiac-25
Ours	$\approx 30\text{M}$	0.99%	our 2.1M split

Table 2: Protocol caveat: test sets differ; numbers are not directly comparable. We release our split for reproducibility.

6 Discussion

Data curation dominates capacity. A $50\times$ smaller encoder matches a 1.5B model once the corpus is cleaned and scaled. We conjecture the remaining 1% DER is dominated by genuinely ambiguous passages (quotations, poetry, archaic orthography).

7 Limitations

Our evaluation uses a held-out split of our own corpus rather than SadeedDiac-25 (access-gated at time of writing). Cross-test-set comparison should be treated as indicative, not definitive.

8 Reproducibility

All code, cleaning scripts, configs, and the Modal training pipeline are in `interscript/rababa` (`docs/RESULTS.md` for exact run IDs).

References

[1] Sadeed: Arabic Diacritization. arXiv:2504.21635, 2025.